

Introduction to Radiobiological Models

A short story



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- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications

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- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

Mathematical Modeling

Poisson Statistics

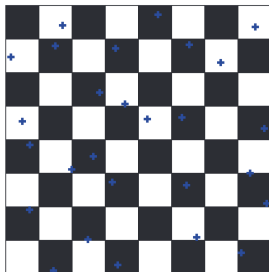


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

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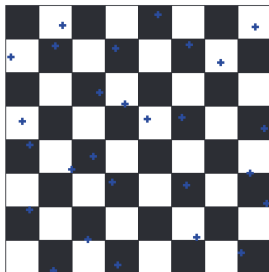


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- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

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- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

Poisson Distribution

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D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

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Note

To achieve a certain TCP, you should aim for a tumor cell survival of $(1 - TCP)$.

Target theory

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where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

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- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

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- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

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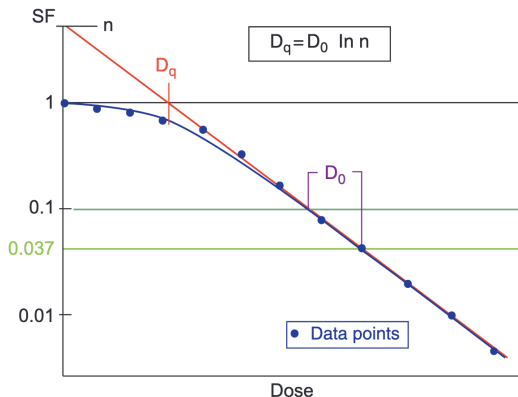


Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

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 - D_q tells you how wide the shoulder is.
 - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

Single-Hit, Multitarget Model

Advantages and Disadvantages

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